

Garbling a Post-Quantum STARK Verifier for Bitcoin

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Abstract

BitVM-style protocols verify succinct proofs on Bitcoin by garbling a proof verifier off-chain. Every garbled verifier and every witness-encryption alternative built so far checks an elliptic-curve SNARK, which a quantum adversary breaks. A hash-based STARK verifier avoids this but is hard to garble: it branches on data, its transcript is defined only by an implementation, hashing dominates its cost, and a stored circuit of 10^8 gates does not fit in memory. We garble Plonky3’s complete multi-STARK verifier over WHIR, with all arithmetic in the binary tower field $\mathbb{F}_{2^{128}}$ and Blake3 for commitments, transcript and garbling. Additions are free under free-XOR, a multiplication costs 2,187 AND gates, a Blake3 compression 10,281, and a streaming garbler holds one byte per wire and a label per live wire. For a Keccak- f trace of 2^{18} rows at 104 bits of soundness, the verifier has 94.1M non-free gates and garbles to 1.5 GB in under a minute on one core: about $30\times$ fewer non-free gates than the Boolean-garbled Groth16 verifier, with no pairing and no trusted setup. We place it in a BitVM3-style protocol on today’s Bitcoin, where a dispute publishes the proof’s million bits: 2.5 MvB with adaptor signatures, 12 to 17 MvB with hash-based keys, and a fifth less after tuning the proof. A quantum analysis shows that the parameters offer about 52 bits against a quantum adversary, that the 128-bit field caps any configuration near 54, and that 100 quantum bits need 256-bit labels and field, for an estimated $3\times$ the gates.

1 Introduction

Bitcoin cannot verify a succinct proof in Script, but a trust-minimized bridge needs exactly that. Consider a withdrawal from a rollup: an operator has paid out a user on Bitcoin and must prove, to a Bitcoin script, that the rollup’s state authorizes the payment; the proof is a STARK over the rollup’s execution. Hashing dominates such statements, and we evaluate on a hash-heavy standard case, proofs of Keccak- f permutations [Nat15].

A SNARK verifier written in Script takes hundreds of megabytes [GKS⁺26]. The BitVM line of work therefore verifies *optimistically*: a prover claims a result, and a challenger disputes it only if it is wrong. BitVM2 [LAA⁺25] settles a dispute by bisecting the verifier’s execution trace on-chain, which is secure but expensive: a mainnet experiment put the unhappy path at over \$14,000 [GKS⁺26]. BitVM3 moves the verifier off-chain into a *garbled circuit* [Lin25, WAA⁺26, Che25]. At setup the verifier circuit is garbled. At dispute time the proof is published under one-time signatures that give the evaluator the circuit’s input labels, and the outcome depends on which output label the evaluation yields. On-chain, only the signed proof and a hashlock remain, at tens of dollars [GKS⁺26].

The off-chain side is where the existing designs fall short. A garbled circuit has to be generated, stored for the lifetime of the deposit and checked, because the garbler may cheat, typically by cut-and-choose over many instances. The Boolean-garbled Groth16 verifier [Bit25] has about 3×10^9 non-free gates, some 42 GiB, and hours of cut-and-choose setup [GKS⁺26]. Garbling Groth16 with native group operations [KFLT26] and witness encryption [GKS⁺26] shrink the artifact to megabytes. All of these designs, however, verify an elliptic-curve SNARK, pairing-based or its

designated-verifier form over BN254 or a binary curve: a quantum adversary breaks them, and they need a trusted or secret setup. A STARK proof must first be wrapped in such a SNARK, which reintroduces both.

Key idea. We garble a hash-based STARK verifier directly and make it cheap by choosing its field. Under free-XOR [KS08] a garbled circuit pays only for non-linear gates, and in a field of characteristic two every addition is an XOR. A STARK over the binary tower field $\mathbb{F}_{2^{128}}$ with Blake3 commitments therefore costs, as a garbled circuit, little more than its hashes and a moderate number of field multiplications. Its soundness, commitments, transcript and garbling PRF all rest on Blake3.

Challenges. Four problems stand between this idea and a working garbled verifier.

- C1 *A fixed circuit for a branching verifier.* A garbled circuit has one shape for all inputs, while the verifier redraws challenges and fixes coordinates depending on sampled values.
- C2 *A transcript defined only by code.* The circuit must absorb and sample exactly the bytes that the prover’s implementation does, or Fiat–Shamir soundness is lost, and that byte schedule is specified nowhere outside the implementation.
- C3 *Hashing dominates.* Merkle paths, leaf rows and the transcript are Blake3 compressions, and a compression costs more than four field multiplications.
- C4 *Memory.* A verifier of 10^8 non-free gates has several hundred million wires; a builder that stores the gate list needs tens of gigabytes before garbling starts.

Contributions. The first three contributions answer C1–C4; the last three place the verifier on Bitcoin and measure it.

- **A fixed-shape circuit for the full verifier (C1, C2).** We implement Plonky3’s multi-STARK verifier [Plo26] over the Boolean WHIR trace commitment as a Boolean circuit with one output wire: the transcript, the zerocheck of the AIR, the batching of the opened columns, the bit ring switch and the WHIR opening. Data-dependent branches are fixed and guarded, and the transcript follows a byte schedule recovered from Plonky3’s own verifier, so one circuit serves every proof of a configuration and can be garbled before the proof exists (§3).
- **Cheaper gadgets (C3).** Tower-field multiplication at 3^k AND gates per 2^k bits, with squaring and multiplication by the tower generator free, and a Blake3 circuit at 10,281 AND gates per compression instead of 20,657, with tree mode for long inputs (§3.1).
- **A streaming garbler (C4).** A plan pass counts each wire’s uses, and the garbling pass garbles and checks each gate as the builder emits it and releases a wire’s slot after its last use (§4).
- **A protocol on today’s Bitcoin and its cost.** The verifier in a BitVM3-style dispute with cut-and-choose, three ways to publish the proof on chain, and a sweep of the proof’s parameters that bounds how far its input shrinks (§6). This is where a STARK pays: a dispute publishes the million-bit proof, 2.5 MvB with adaptor signatures and 12 to 17 MvB with hash-based keys, against a few kilobytes for Groth16, and tuning removes only a fifth of it.

- **A post-quantum analysis.** Bounds for every component against a quantum adversary, from Fiat–Shamir in the quantum random-oracle model to Grover search on the garbling offset, and the parameters and cost that restore a quantum security target (§5.1).
- **Measurements.** Real Keccak- f proofs from 2^5 to 2^{20} rows, a gate profile, an ablation of the gadget and memory choices, the on-chain cost of a dispute, and a comparison with pairing-based designs (§7).

The circuit builder and the measurements are open source; the streaming garbler, the Blake3 circuit and a label-level evaluator are contributed to GOAT’s bitvm-gc [GOA26b].

2 Setting

2.1 Garbled verifiers on Bitcoin

We follow the BitVM3 setting [WAA⁺26]. A *garbler*, the operator who proves, and an *evaluator*, a challenger, agree on a verifier circuit C with one output wire, and C is garbled at setup. At dispute time the operator publishes its proof on-chain under one-time keys, and each published bit gives the evaluator the input label of that bit’s value. The proof is public, so we use *privacy-free* garbling: the evaluator may learn every wire’s value. What garbling must provide is *authenticity*. Holding the labels of an input x , the evaluator obtains the label of the output value $C(x)$, and it cannot obtain the label of the other value. A hashlock on the label of value 0 lets a challenger who evaluates an invalid proof stop the operator’s claim.

The garbler may cheat, so the garbling is checked by cut-and-choose, as in BABE [GKS⁺26], Mosaic [KTA⁺26] and bitvm-gc [GOA26b]; its cost is linear in the size of the garbled circuit, which this paper reduces. Section 6 gives the full protocol and its on-chain cost.

2.2 The garbling scheme

We use privacy-free garbling [FNO15] with free-XOR [KS08] and one ciphertext per non-free gate, in the syntax of Bellare, Hoang and Rogaway [BHR12b].

Let Δ be a random secret 128-bit offset. Every wire w has a false label w_0 , and its true label is $w_1 = w_0 \oplus \Delta$. Let $H(\ell, g) = \text{Blake3}(\ell \parallel g)$ truncated to 128 bits, where g is the gate’s index. An XOR gate sets $c_0 = a_0 \oplus b_0$ and has no ciphertext. An AND gate $c = a \wedge b$ sets

$$c_0 = H(a_0, g), \quad ct = H(a_1, g) \oplus H(a_0, g) \oplus b_0.$$

An evaluator holding the label a of value x , and b of value y , outputs $H(a, g)$ if $x = 0$ and $H(a, g) \oplus ct \oplus b$ if $x = 1$. In both cases this is the label of $x \wedge y$. An OR gate is symmetric, with $c_0 = H(a_1, g) \oplus \Delta$ and $ct = H(a_1, g) \oplus H(a_0, g) \oplus b_1$. The garbled circuit is one 16-byte ciphertext per AND or OR gate.

Two details matter in practice. First, Δ must be secret and fresh for each garbling: with a public Δ any label yields its complement, and the output’s other label is free to compute. Second, a hash-based PRF gives labels no algebraic structure. RSA-based garbling proposals for BitVM3 were broken exactly through such structure: label forgery by extended Euclid on coprime exponents, and by the Franklin–Reiter related-message attack [FLB25].

2.3 The proof system

The proof system is Plonky3’s multi-STARK [Plo26] over the Boolean WHIR trace commitment, with every field in the Wiedemann tower [Wie88]

$$\mathbb{T}_0 = \mathbb{F}_2, \quad \mathbb{T}_{k+1} = \mathbb{T}_k[X_k]/(X_k^2 + X_{k-1}X_k + 1), \quad X_{-1} = 1,$$

up to $\mathbb{T}_7 = \mathbb{F}_{2^{128}}$. An element of $\mathbb{T}_{k+1}$ is stored as its low half (the coefficient of 1) and its high half (the coefficient of X_k), recursively down to bits. The trace is committed as bits: a Boolean column is packed into $\mathbb{F}_{2^{128}}$ elements, and WHIR commits to the packed multilinear polynomial on an additive (Cantor) evaluation domain [Can89, LCH14]. Binary towers and ring switching come from Binius [DP25, DP24].

Terms. An *AIR* (algebraic intermediate representation) is a set of polynomial constraints over a trace’s current and next rows. A *zerocheck* [CBBZ23] proves that the constraints vanish on every row by a sumcheck [LFKN92] against the equality polynomial $\text{eq}(\tau, \cdot)$ at a random point τ . *WHIR* [ACFY25] is a Reed–Solomon proximity test used as a multilinear polynomial commitment. Its *STIR queries* [ACFY24] open the committed, folded codeword at random positions, and its *out-of-domain* (OOD) samples bind the committed polynomial at points outside the evaluation domain. The *Cantor basis* is the $\mathbb{F}_2$ -basis $v_0 = 1$, $v_i^2 + v_i = v_{i-1}$, which gives binary fields an additive FFT domain. The *bit ring switch* turns a claim about a polynomial’s bits into a claim about its packed $\mathbb{F}_{2^{128}}$ representation. *Johnson list-decoding regime* refers to the proximity parameters for which the soundness bounds are proven via the Johnson bound, without conjectures.

Verification. The verifier:

1. runs a *zerocheck* of the AIR: a degree-4 sumcheck over the row variables, then the AIR constraints, evaluated at the bound point from the opened current and next rows and Horner-batched under a challenge α , against $\text{eq}(\tau, r)$;
2. batches the opened columns into one bit-level claim, at a column point drawn from the transcript;
3. runs the *bit ring switch*, which turns that claim into an $\mathbb{F}_{2^{128}}$ claim on the packed polynomial. It uses a 128×128 tensor and, above 2^7 rows, its carry and last tensors, a batched sumcheck, and a closing weight computed in the tensor algebra;
4. runs the WHIR opening of that claim: sumcheck rounds with proof of work, STIR queries on Merkle-committed folded codewords, and a final polynomial.

All challenges come from a Blake3 [OANWO20] sponge over bytes, and all Merkle trees use Blake3 compressions.

3 The verifier as a circuit

This section turns the verifier of §2 into a Boolean circuit with one output wire. Figure 1 shows its stages in transcript order, with each stage’s share of the non-free gates of our running example: a 2^{18} -row Keccak- f trace with 1,625 columns, 1,650 AIR constraints and 22 packed variables, whose proof is 140 KiB and whose verifier has 94.1M non-free gates. We first describe the gadgets, then the transcript, the multi-STARK layers and WHIR.

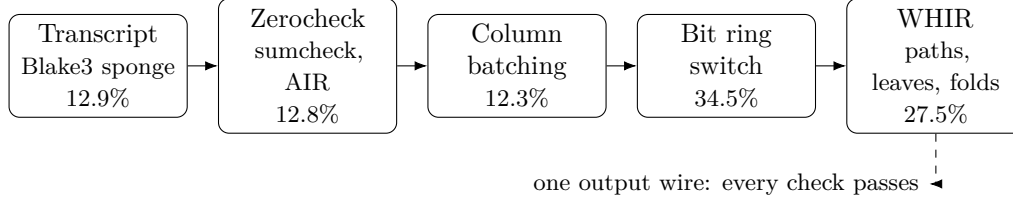


Figure 1: The ring switch dominates the verifier circuit, followed by WHIR. The stages appear in transcript order, with each stage’s share of the 94.1M non-free gates of the 2^{18} -row verifier: transcript is the STARK and WHIR sponges, zerocheck is its sumcheck, AIR constraints and closing, and WHIR is everything after the ring switch except its transcript.

gadget	AND gates
$\mathbb{F}_{2^{128}}$ multiplication	2,187
$\mathbb{F}_{2^{128}}$ squaring, multiplication by X_{k-1} , addition	0
Blake3 of 64 bytes (a Merkle compression)	10,281
Blake3 of 256 bytes (a leaf row of 16 elements)	41,511
Blake3 of 1,072 bytes (two chunks: a transcript flush)	186,125

Table 1: Gadget costs, checked against `p3-binary-field` and the `blake3` crate.

3.1 Gadgets

Tower arithmetic. A wire vector for an element of $\mathbb{T}_k$ is its bit pattern, so gadget outputs can be read back directly as field elements. Multiplication follows the tower recursively, Karatsuba at each level:

$$z_0 = a_0 b_0, \quad z_2 = a_1 b_1, \quad z_1 = (a_0 + a_1)(b_0 + b_1) + z_0 + z_2, \quad ab = (z_0 + z_2) + (z_1 + X_{k-1} z_2) X_k.$$

Additions are free. Multiplication by the generator X_{k-1} is a rewiring plus XORs, and squaring is linear over $\mathbb{F}_2$. The only AND gates are the base-field products, three per level, so a 2^k -bit multiplication costs 3^k AND gates: 2,187 in $\mathbb{F}_{2^{128}}$. A multiplication by a constant folds to XORs entirely.

Blake3. We start from `bitvm-gc`’s Blake3 circuit and change two things. A 32-bit addition’s carry is the majority

$$\text{maj}(a, b, c) = c \oplus ((a \oplus c) \wedge (b \oplus c)),$$

one AND per bit, and the carry out of the top bit is discarded, so it is not computed. This halves a compression from 20,657 to 10,281 AND gates. We also add Blake3’s tree mode (chunk chaining values on a stack, parent nodes, the root last). The verifier’s transcript absorbs more than one 1,024-byte chunk between samples: the final polynomial is 64 coefficients, 1,024 bytes on its own. Table 1 gives the costs. Every gadget is checked against the reference implementation (`p3-binary-field`, the `blake3` crate) on random inputs.

3.2 Transcript

The transcript is Plonky3’s challenger over a Blake3 byte sponge. Observing appends to an input buffer; sampling, when the output buffer is empty, hashes the input buffer, replaces it with the digest, and pops bytes from the digest’s end. A field element is 16 little-endian bytes. A query

index is 8 bytes read as a little-endian integer and masked. A proof-of-work check absorbs the witness and requires a fixed number of sampled bits to be zero. We implement the same challenger on wires. Its schedule has no data-dependent branch: queries are a fixed number of masked draws, and a proof-of-work check a fixed number of bits. So the circuit’s shape is a function of the configuration alone. The sub-transcripts’ domain separators are constants, baked in as constant wires, and everything the prover sends is an input.

The byte schedule, that is, the order in which roots, out-of-domain answers, sumcheck messages, proof-of-work witnesses, stratified query draws and the final polynomial are absorbed and sampled, is implicit in Plonky3’s implementation and not specified separately. We recovered it by running Plonky3’s own verifier through a challenger that logs every byte, and we test the circuit against that log op for op.

3.3 The multi-STARK layers

The zerocheck, the column batching and the ring switch are implemented gate for gate from a reference implementation that is checked op for op against Plonky3’s verifier. The AIR arrives as Plonky3’s symbolic constraints and is evaluated on wires.

Two of Plonky3’s data-dependent branches cannot appear in a fixed circuit. We fix them and guard them with checks.

- Plonky3 redraws a zerocheck challenge coordinate that happens to be zero. The circuit takes a fixed number of draws and requires each to be nonzero.
- Plonky3’s ring switch would fix a column-point coordinate that is exactly 0 or 1. The circuit assumes there is no such coordinate and checks it.

Each event has probability about 2^{-127} per coordinate. When it occurs the circuit rejects where Plonky3 would continue, never the reverse.

3.4 WHIR

The WHIR layer checks every proof-of-work witness. It hashes every opened row and walks it up its Merkle path to the root absorbed in the transcript. It then folds each queried row with the round’s folding randomness, and checks the final queries against the final polynomial and the closing sumcheck identity. Plonky3 sends Merkle openings pruned: siblings that another query can recompute are left out. A fixed circuit cannot follow a data-dependent frontier, so the proof is expanded outside the circuit into one full path per query. This costs back the pruning’s savings, and it is the right trade for a circuit whose shape must not depend on the proof. When the commitment uses a Merkle cap of 2^h roots, the transcript absorbs all of them, and each query selects its root with a multiplexer on the top h index bits: $256(2^h - 1)$ AND gates, against h compressions saved.

4 Streaming garbling

The circuits here have 10^8 – 10^9 non-free gates and several times as many wires. A builder that stores the gate list, as bitvm-gc’s `CircuitAdapter` does, needs tens of gigabytes before garbling starts. Upstream’s own DV-SNARK verifier test, for example, exceeds 64 GB. Nothing in garbling needs the gate list, though: a gate’s ciphertext depends only on its input labels.

Our garbler is a circuit-builder backend that garbles each gate as it is emitted. It uses two passes of the same deterministic builder (Algorithm 1). The *plan* pass records, for each wire by creation

Algorithm 1 Streaming garbling of a deterministic builder $\mathcal{B}$

```
1: plan: run  $\mathcal{B}$  on a backend that only counts:  $uses[k] \leftarrow$  reads of the  $k$ -th created wire
2: pin the constants and the outputs:  $uses[k] \leftarrow \infty$ 
3: garble:  $\Delta \leftarrow \{0, 1\}^{128}$ ; run  $\mathcal{B}$  again on the backend:
4:   on input wire  $k$ :  $\ell_0[s] \leftarrow \text{Blake3}_{seed}(k)$  in a free slot  $s$ ;  $v[s] \leftarrow$  its value
5:   on XOR( $a, b$ ):  $\ell_0[s] \leftarrow \ell_0[a] \oplus \ell_0[b]$ ;  $v[s] \leftarrow v[a] \oplus v[b]$ 
6:   on AND( $a, b$ ) with gate index  $g$ :
7:      $(c_0, ct) \leftarrow \text{Garble}(\ell_0[a], \ell_0[b], g, \Delta)$ ; emit  $ct$ 
8:      $e \leftarrow \text{Eval}(\ell_{v[a]}[a], \ell_{v[b]}[b], ct, g, v[a])$ 
9:      $\ell_0[s] \leftarrow c_0$ ;  $v[s] \leftarrow v[a] \wedge v[b]$ ; assert  $e = \ell_{v[s]}[s]$ 
10:  on OR( $a, b$ ): as for AND, with the OR garbling of §2.2
11:  after each read of  $a$ : decrement  $uses[a]$ ; release  $a$ 's slot at zero
```

index, the number of gates that read it: one byte per wire, with 255 meaning “never release”. The *garbling* pass hands out slots. A wire’s slot is released after its last read, and a later wire takes it. Wire identifiers are slots, which the builder interface permits. The two constant wires, and any wire read after the build (an output), are pinned. Both passes apply the same constant folding ($x \oplus x = 0$, $x \wedge 1 = x$, $\dots$), so they see the same wires. Slot storage is chunked, so that a growing array never reallocates.

When the garbler holds a witness, it evaluates as it garbles. At each non-free gate it runs the evaluator’s formula on the labels of the actual values, and requires the result to equal the label of the gate’s output value. At setup no proof exists, and none is needed: garbling a gate uses only its input labels, so the garbler runs the garbling pass with every input held at 0 and keeps the ciphertexts. A challenger later evaluates them with an evaluator backend that runs the same builder on the proof and holds one label per wire. We tested exactly this. Garbled with every input at 0 and then evaluated, the stored garbling gave the label of 1 on a real proof and the label of 0 on the same proof with one opened value changed, at 2^5 and 2^8 rows. Evaluation took 11.5s and 17.0s, against 20.6s and 34.0s for garbling, since it hashes once per non-free gate where garbling hashes twice. Memory is one byte per wire for the plan, 0.61 GiB for the 659M wires at 2^{18} rows, plus 18 bytes per live wire (a 16-byte label, the value and a use counter), 41 MB at the peak of 2.26M live wires. The plan depends only on the circuit, so every garbled instance of a configuration can share it.

5 Security

This section states what the garbled verifier guarantees and on which assumptions. It combines the soundness of the proof system, the effect of the two guarded branches, and the authenticity of the garbling.

The circuit and the verifier. When neither guarded event of §3 occurs, the circuit computes exactly Plonky3’s verifier; we test this op for op against Plonky3’s logged transcript and on real proofs. When a guarded event occurs, the circuit outputs 0. Hence an output of 1 implies that Plonky3’s verifier accepts, and an honest proof is rejected only if a guarded event occurs, with probability at most $\varepsilon_{\text{guard}} \approx n \cdot 2^{-127}$ for n guarded coordinates.

trace rows	2^5	2^8	2^{12}	2^{16}	2^{18}	2^{20}
composed soundness (bits)	106.1	105.2	104.7	104.3	104.3	104.0

Table 2: Soundness stays at about 104 bits at every measured size, 103.95 at 2^{20} rows. The values are Plonky3’s composed soundness for each configuration.

Soundness of the proof system. We use Plonky3’s own security assessment of each configuration. Every term is a proven bound, the proximity terms in the Johnson list-decoding regime, and nothing is left unassessed (Table 2). At 2^{18} rows the binding term is the WHIR opening at 104.3 bits (110 bits per term, rate $1/32$, folding factor 4, proof of work up to 32 bits). The other terms are constraint batching 109.0, zerocheck 114.5 and its sumcheck 113.5, bit ring switch 114.0, column batching 115.2, and commitment and transcript collisions 128 (Blake3). We write $\varepsilon_{\text{STARK}}(t)$ for the composed error against an adversary making t hash queries. Plonky3’s report states it as bits of security, 104.3 at 2^{18} rows, meaning that $\varepsilon_{\text{STARK}}(t) \leq t \cdot 2^{-104.3}$ for a classical adversary; §5.1 gives the quantum bound.

Authenticity of the garbling. Our scheme is privacy-free garbling [FNO15] with free-XOR [KS08] and one ciphertext per non-free gate [ZRE15], with $H = \text{Blake3}$ and a fresh secret Δ per garbling. Free-XOR is secure when H is circular correlation robust [CKKZ12]; we assume this of Blake3, as is standard for hash-based garbling. In the BitVM setting the garbled circuit is published at setup and the proof is chosen afterwards, so authenticity must hold for inputs chosen adaptively, after the garbled circuit is seen [BHR12a]. Let ε_{GC} be the advantage of an evaluator that sees the garbled circuit, then chooses an input x , receives its labels, and outputs the label of the value $\neg C(x)$ on the output wire. We assume that our Blake3-based free-XOR garbling has adaptive authenticity in this sense, with Blake3 modelled as a random oracle; Bellare, Hoang and Rogaway give constructions proven adaptively secure [BHR12a].

Theorem 1. *Let C be a correctly garbled verifier of a fixed configuration, and x the encoding of a proof π for a statement.*

- (i) *Soundness. If the statement is false, an evaluator holding the labels of x obtains the output label of value 0, except with probability at most $\varepsilon_{\text{STARK}}(t)$, where t bounds the hash queries of the party that chose π .*
- (ii) *Authenticity. If π is honestly generated, an evaluator holding the labels of x obtains the output label of value 0 with probability at most $\varepsilon_{\text{guard}} + \varepsilon_{\text{GC}}$.*

Proof sketch. (i) If $C(x) = 1$, Plonky3’s verifier accepts π by the previous paragraphs, which for a false statement happens with probability at most $\varepsilon_{\text{STARK}}(t)$. Otherwise $C(x) = 0$, and correctness of the garbling yields the label of value 0. (ii) For an honest proof $C(x) = 1$ unless a guarded event occurs, and the label of value 0 is then the label of $\neg C(x)$, which the evaluator obtains with probability at most ε_{GC} . $\square$

Section 6 uses (i) to protect challengers against a false claim and (ii) to protect an honest operator.

5.1 Post-quantum analysis

No step of the construction uses a number-theoretic assumption: the proof system, the commitments, the transcript and the garbling rely on Blake3 alone. Pairing-based verifiers (Groth16, Pari

and their designated-verifier variants), the witness-encryption schemes built on them (§8) and Bitcoin’s own Schnorr signatures fall to Shor’s algorithm [Sho94]. Our verifier does not. Relying on hashing alone is necessary for post-quantum security but not sufficient [ZKM26]: a quantum adversary still speeds up generic search over hash inputs. This subsection bounds that speed-up for every component and gives parameters that restore a target level. Table 3 summarises it.

Fiat–Shamir against quantum adversaries. In the quantum random-oracle model, Chiesa, Manohar and Spooner [CMS19] show that compiling a public-coin interactive oracle proof with round-by-round soundness error ε_{rbr} with a hash bounds the success of a quantum adversary that makes t hash queries by $O(t^2\varepsilon_{\text{rbr}} + t^3/2^\lambda)$, where $\lambda = 256$ is the digest length. A classical adversary is bounded by $O(t\varepsilon_{\text{rbr}})$.

WHIR has round-by-round soundness [ACFY25, §5]. For the sumcheck-based layers (zerocheck, column batching, ring switch) we assume it, as is standard for sumcheck, with each of Plonky3’s terms as the round error. A round with error 2^{-a} therefore offers about $a/2$ bits against a quantum adversary, and Grover search [Gro96] attains this generically when the bad challenges form an unstructured set. Proof-of-work grinding is generic search too: b bits of grinding cost 2^b hashes classically and about $2^{b/2}$ quantumly.

The proof system at our parameters. At 2^{18} rows each WHIR term targets 110 bits, of which grinding supplies 26 to 32 bits (per round: 33 queries with 30 and 28 bits of grinding, 20 with 32 and 30, 15 with 29 and 32, and 12 terminal queries with 27). Both parts halve under quantum search, so each WHIR term offers about 55 bits, and the composed 104.3 bits become about 52. The algebraic terms (zerocheck 114.5, its sumcheck 113.5, ring switch 114.0, column batching 115.2, constraint batching 109.0) are Schwartz–Zippel errors over the 2^{128} -element field and offer 54 to 58 bits. They are capped by the field: no number of queries raises them. Collisions in the 256-bit Blake3 digests cost about 2^{85} quantum queries [BHT98], and preimages 2^{128} .

The ceiling of the current configuration space. Raising Plonky3’s per-term target does not raise the quantum level. We searched the target that the proven-list-decoding profile accepts at 2^{18} rows. Up to 134 bits per term it keeps the queries at $33 + 20 + 15 + 12$ and raises the grinding to 51–56 bits, its ceiling; at 200 bits it refuses, needing 103 bits of grinding. Plonky3’s security report fully assesses targets up to 117 bits per term, where the composed soundness is 108.6 bits, bound by constraint batching at 109.0, a field-size term no WHIR parameter moves. Grinding and the field-bound terms both halve under quantum search, so no configuration of the current proof system exceeds about 54 quantum bits. Reaching a quantum target is a change to the proof system, not to its parameters.

The garbling. Authenticity rests on the secrecy of Δ , and Δ can be tested against a single gate. An evaluator holding the labels a, b of an AND gate with known values x, y and its ciphertext ct checks a candidate Δ' by recomputing ct from $a, a \oplus \Delta', b$ and Δ' . A wrong candidate passes with probability 2^{-128} . Grover search therefore recovers Δ with about 2^{64} quantum evaluations of Blake3, after which every label’s complement is known and the output label of the other value is free. With 128-bit labels the garbling offers about 64 bits against a quantum adversary. The same holds for a hashlock on a 128-bit output label: a preimage search over 2^{128} labels costs about 2^{64} . Grover search is highly sequential, and the adversary has only the time between setup and the end of the dispute window, but we do not rely on this.

component	attack	classical	quantum, as measured	to reach λ_Q quantum bits
WHIR opening	FS search, grinding	104.3 (max. 111.3)	≈ 52	$\approx 2\lambda_Q$ per term: $\approx 2.1 \times$ queries at $\lambda_Q = 100$
zerocheck, sumchecks, ring switch	FS search	109–115	≈ 54 –58	a field of $\gtrsim 2^{2\lambda_Q+14}$ elements: $\mathbb{F}_{2^{256}}$
Merkle and transcript hash	collision, preimage	128	≈ 85	unchanged (256-bit Blake3)
garbling (Δ , output hashlock)	Grover on 128-bit labels	128	≈ 64	$2\lambda_Q$ -bit labels: 256 bits
hash-based input keys	preimage of HASH160, 128-bit preimages	128	≈ 64	20-byte preimages (≈ 80); a 256-bit hash beyond
adaptor keys, pre-signed graph	Shor	–	broken	hash-based keys; post-quantum signatures in Bitcoin

Table 3: Against a quantum adversary the current parameters offer about 52 bits, half their classical level; restoring 100 bits needs more queries, a 256-bit field, 256-bit labels and a 256-bit input hash, and signatures that Bitcoin does not currently offer. Bits of security per component, classically and against a quantum adversary under the generic square-root speed-up.

The surrounding protocol. The protocol of §6 publishes input bits under one-time keys. Lamport and Antichain Winternitz keys rest on HASH160 preimages. With our 16-byte preimages a quantum search over the preimages costs about 2^{64} , and 20-byte preimages raise it to 2^{80} for 4 more witness bytes per bit. Schnorr adaptor signatures, the committee’s pre-signed transactions and any Taproot key path fall to Shor’s algorithm. A post-quantum deployment therefore needs hash-based input keys, which the protocol supports at 5 to 7× the on-chain cost of adaptor signatures (Table 7), and post-quantum signatures in Bitcoin itself for the pre-signed graph, which no current opcode provides.

Parameters for a quantum target. To offer λ_Q bits against a quantum adversary, every classical term must reach about $2\lambda_Q$ bits. Three changes follow.

- The WHIR terms need about $2\lambda_Q$ bits each. Since grinding also halves, the query count must carry most of it: from about 80 information-theoretic bits per term today to about 168 at $\lambda_Q = 100$, about $2.1 \times$ the queries. Plonky3’s profile raises grinding instead, so this needs a query-heavy profile.
- The algebraic terms need challenges from a field of about $2^{2\lambda_Q+14}$ elements, which means one more tower level, $\mathbb{F}_{2^{256}}$, at $3^8 = 6,561$ AND gates per multiplication, three times today’s. Repeating each algebraic check with independent challenges is an alternative at a similar cost.
- Labels of $2\lambda_Q$ bits: 256-bit labels double the ciphertext size to 32 bytes per non-free gate. The gate count is unchanged, because labels feed the garbling hash, not the circuit.

Applying these factors to the measured 2^{18} profile gives an estimate, not a measurement. Merkle paths and leaf hashes scale with the query count ($2.1 \times$), and leaves also with the element size ($2 \times$), to about 45M gates. The transcripts absorb twice the bytes, about 24M. Multiplications triple, and those that depend on the queries also scale with them, to about 213M. That is about 280M non-free gates, $3 \times$ today’s, and about 9 GB garbled at 32 bytes per gate: about $10 \times$ fewer non-free gates than the Boolean-garbled Groth16 verifier, which offers no quantum security at all.

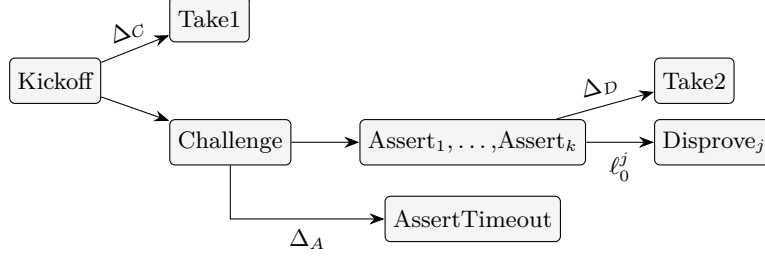


Figure 2: The dispute costs one challenge and the operator’s publication of the proof; the happy path costs two constant-size transactions. Pre-signed transactions of one claim. Labels on edges are relative timelocks, and Disprove_j spends a hashlock on $h_j = \text{SHA256}(\ell_0^j)$, one leaf per kept instance j . Take1 and Take2 pay the operator; Disprove and AssertTimeout burn its stake, with a reward to the challenger.

6 Verifying on Bitcoin

This section places the garbled verifier in a protocol that today’s Bitcoin enforces, without a soft fork or a data-availability assumption, and prices it. We take BitVM3’s roles [WAA⁺26, KTA⁺26]: the operator, who is the prover, garbles, and challengers evaluate. The running application is a bridge: the operator has paid a withdrawal from its own funds and claims reimbursement from a deposit, backed by a stake that it loses if a challenger shows the claim false.

Setup. The operator garbles N instances of C from independent seeds. A challenger chooses $N - M$ of them to open, re-garbles each opened instance from its seed and compares. The streaming garbler does this in one garbling pass per instance, reusing the plan that all instances share (§4). The challenger stores the M kept instances, 1.5 GB each at 2^{18} rows. For each kept instance j the operator commits $h_j = \text{SHA256}(\ell_0^j)$, the hash of the label of output value 0 (reject). It also fixes one-time keys for the input bits of C , and the kept instances’ input labels are *soldered* to them: publishing bit b of input i under the keys gives a challenger the label of value b on wire i in every kept instance, and no other label. The transactions below are pre-signed at setup by a committee, as in BitVM2 [LAA⁺25]; as there, their enforcement needs one honest signer.

Transactions. Figure 2 shows the graph of one claim.

- *Kickoff*: the operator claims reimbursement for a withdrawal it has paid. If no challenge follows within Δ_C , *Take1* pays it. This is the happy path.
- *Challenge*: any challenger may spend *Kickoff*’s challenge output, paying the operator a bond that covers the cost of *Assert*, as BitVM2’s crowd-funded challenge does [LAA⁺25].
- *Assert₁, ..., Assert_k*: the operator publishes the bits of its proof under the one-time keys. The proof is a million bits (Table 4), so *Assert* is split into k standard transactions. If *Assert* is not complete within Δ_A , *AssertTimeout* burns the stake.
- *Disprove_j*: within Δ_D of the last *Assert*, a challenger that holds ℓ_0^j for some kept instance j spends the *Assert* output through the leaf `OP_SHA256  $h_j$  OP_EQUAL`, which burns the stake and makes *Take2* unspendable. Otherwise *Take2* pays the operator after Δ_D .

The statement’s public inputs, such as the rollup state that authorizes the withdrawal, are committed in Kickoff under the same kind of keys and are input bits of C ; the Keccak- f AIR we measure has none.

Dispute. A challenger reads the published bits from the chain, obtains through the soldering the input labels of every kept instance, and evaluates each instance. The evaluation takes one Blake3 call per non-free gate, half of what garbling takes, and runs off-chain: 17 s at 2^8 rows (§4). If the proof is invalid, a correctly garbled instance yields ℓ_0^j , and the challenger publishes Disprove_j , a hashlock spend of constant size. Nothing on-chain depends on the size of C ; only Assert depends on the size of the proof.

Security. Let ε_{cc} bound the probability that every kept instance is malformed while every opened one is correct; for choosing $N - M$ of N uniformly this is $1/\binom{N}{M}$, which is 2^{-40} at $(N, M) = (181, 7)$ [GKS⁺26]. Let ε_{sol} bound a failure of the soldering, and ε_{sig} a forgery of the one-time keys. Theorem 1 then gives two properties, assuming the committee’s graph is enforced and transactions confirm within their timelocks.

- *A false claim is disproved.* If the operator’s statement is false, an honest challenger obtains some ℓ_0^j except with probability $\varepsilon_{\text{STARK}}(t) + \varepsilon_{cc} + \varepsilon_{sol}$: a correctly garbled kept instance exists, its labels are those of the published proof, and Theorem 1(i) applies. An operator that publishes no proof loses its stake to AssertTimeout.
- *An honest operator keeps its stake.* If the operator’s proof is honest, no challenger obtains any ℓ_0^j except with probability $M(\varepsilon_{\text{guard}} + \varepsilon_{GC}) + \varepsilon_{sol} + \varepsilon_{sig}$, by Theorem 1(ii) and a union bound over the kept instances.

This is a sketch, not a proof in a ledger model; BitVM3 [WAA⁺26] and BABE [GKS⁺26] give such proofs for their instantiations, whose structure ours follows.

Publishing the input. Three one-time schemes can carry the input bits, and they trade on-chain size against assumptions.

- *Lamport.* Each bit has two hashed 16-byte preimages. The leaf script per bit,

OP_HASH160 OP_DUP h_0 OP_EQUAL OP_SWAP h_1 OP_EQUAL OP_BOOLOR OP_VERIFY

is 49 bytes, and the witness 17 bytes. We executed it with 998 bits per input, the most that the 1,000-item stack limit allows, and it rejected wrong preimages. With per-input overhead this is 66.2 weight units, or 16.55 vB, per bit.

- *Antichain Winternitz.* Plain Winternitz chains are unsafe here: from a published chain value the evaluator hashes forward to the labels of other digit values [TF26]. Antichain Winternitz closes this and costs 11.31 vB per bit at parameters (2, 16), with 0.256 kB of off-chain table per bit; its larger parameters save more on-chain but need up to 43.8 kB per bit off-chain [TF26], 46 GB for our input.
- *Schnorr adaptor signatures* over 8-bit digits, as in BitVM3 and Mosaic: 9,707 weight units for a 1,024-bit proof [WAA⁺26], or 2.37 vB per bit. Mosaic solders all kept instances to one set of signatures by polynomial label correlation, with an on-chain footprint independent of the number of instances [KTA⁺26]. The signatures rest on discrete logarithms, and an evaluator that holds the signing key can spend without completing the adaptor [TF26].

Our labels are free-XOR correlated, so a hash-based preimage s cannot itself be a label. It is mapped to one by a translation ciphertext $\text{KDF}(s) \oplus w_b$, two per input bit and instance: 32 bytes per bit, 33 MB per instance at 2^{18} rows, off-chain. Soldering hash-based keys to M kept instances without publishing M sets needs a proof that the opened and kept translations agree, as BABE’s zero-knowledge soldering [GKS⁺26] or Glock’s verifiable secret sharing [Eag25] provide for smaller inputs. We have not built it for a million input bits, and its cost there is open.

An executed dispute. We ran the dispute on a small circuit, with its on-chain steps executed as Bitcoin Script by `bitcoin-scriptexec`. The statement is knowledge of a 64-byte x with $\text{Blake3}(x) = h$: one Blake3 compression, 512 input bits and 10,536 non-free gates. The operator garbles it at setup, before choosing x , and publishes Lamport hashes, translation ciphertexts, the labels of the two constant wires and SHA256 of the reject label. Its Assert reveal runs through a 25 KB Lamport script under the 1,000-item stack limit. The challenger derives the input labels from the published preimages and evaluates the stored garbling without Δ or any label it was not handed. For a true x it obtains the accept label, which the Disprove hashlock script rejects; for x with one bit changed it obtains the reject label, which opens it. Assert rejects a preimage outside the committed keys. We then ran the same dispute on the Keccak- f STARK verifier at 2^5 rows. Garbled with every input at 0, it reads 586,240 input bits, which the operator reveals through 588 Assert scripts of at most 998 bits each, 28.7 MB of script in all, each executed under the stack limit. The challenger’s evaluation of the stored garbling from the revealed labels gives the accept label on the real proof and the reject label, which opens Disprove, on the proof with one opened value changed. Signatures, timelocks and cut-and-choose are not part of either run.

7 Evaluation

The experiments answer five questions: how the garbled verifier scales with the trace (Table 4, Figure 3), where its gates go (Table 5), how much each design choice contributes (the ablation), what a dispute costs on-chain and how far its input shrinks (Tables 7 and 8), and how it compares with pairing-based designs (Table 9).

Setup. We proved Keccak- f permutations with Plonky3’s 1,625-column characteristic-2 Keccak AIR (booleanity is not constrained, since the commitment is to bits). A byte-logging challenger recorded each run, our reference replayed Plonky3’s verifier byte for byte, and the full circuit was then streamed through the garbler. The parameters were rate 1/32, folding factor 4 and 110 bits per WHIR term. Garbling was single-threaded. Every run up to 2^{18} rows was limited to 32 GB of RAM and 8 cores of an Intel Xeon Platinum 8375C at 2.9 GHz; the 2^{20} run, whose proof took longer to generate, ran on an AMD EPYC 9355 limited to 128 GB and 32 cores.

The circuit’s size depends on the AIR: the transcript, the column combination and the constraint evaluation grow with the number of columns and constraints. All sizes below are therefore for this 1,625-column AIR, which is a demanding case; a Groth16 verifier, by contrast, has the same size for every statement.

Scaling. Table 4 and Figure 3 give the full verifier from 2^5 to 2^{20} rows. Every circuit accepted its proof, and at 2^5 and 2^8 we also checked that it rejected the proof with one opened value changed. Growing the trace 32,768-fold grew the circuit 2.2-fold, because the verifier depends on the trace length only through the logarithmic number of variables.

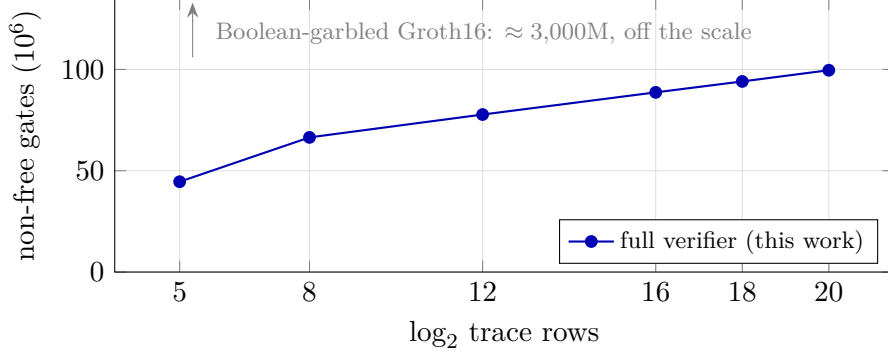


Figure 3: The garbled verifier grows logarithmically with the trace: from 44.6M non-free gates at 2^5 rows to 99.6M at 2^{20} , $2.2\times$ for a $32,768\times$ larger trace. The arrow marks the Boolean-garbled Groth16 verifier, about 3,000M non-free gates, which is off the scale.

trace	vars	non-free gates	garbled	η (B/cell)	garbling	garbler memory	input bits
2^5 rows (1 Keccak- f)	9	44,573,121	713 MB	13,715	35 s	0.32 GiB	586,240
2^8 rows (10)	12	66,459,859	1,063 MB	2,556	53 s	0.46 GiB	743,808
2^{12} rows (163)	16	77,736,394	1,243 MB	186.9	53 s	0.54 GiB	874,240
2^{16} rows (2,621)	20	88,680,747	1,418 MB	13.32	66 s	0.61 GiB	997,760
2^{18} rows (10,485)	22	94,069,548	1,505 MB	3.53	57 s	0.65 GiB	1,041,024
2^{20} rows (41,943)	24	99,613,898	1,593 MB	0.94	42 s [†]	0.68 GiB	1,108,992

Table 4: The full verifier stays below 100M non-free gates and 1.6 GB garbled up to 2^{20} rows, and the garbler needs less than 0.7 GiB. Measured on real Keccak- f proofs; “vars” is the number of packed variables WHIR opens, garbled size is 16 bytes per non-free gate, and η is the garbled size per committed trace cell, 1,625 columns times the rows. Garbling is single-threaded, and its time varied by up to 50% with load on the shared machine (23–35 s at 2^5 across runs). Garbler memory is the plan (one byte per wire) plus 18 bytes per live wire at the peak, computed from the measured counts; at 2^{18} rows the garbler holds at most 2,256,536 labels live. [†]The 2^{20} proof was generated on a second, larger machine, where the circuit was also garbled, again on one core.

Garbling efficiency. To compare garbled verifiers across trace sizes and AIRs, we define the *garbling efficiency* of a configuration as

$$\eta = \frac{|\text{GC}|}{R \cdot W} \quad \text{bytes per committed cell},$$

where $|\text{GC}|$ is the garbled circuit’s size in bytes, 16 per non-free gate, R the number of trace rows and W the number of committed columns. It is a cost per unit of verified computation, so smaller is better. For our Keccak- f AIR, $W = 1,625$ and each cell is one bit. Because $|\text{GC}|$ grows with $\log R$ while $R \cdot W$ grows with R , η falls almost in inverse proportion to the trace: from 13,715 bytes per cell at 2^5 rows to 3.53 at 2^{18} and 0.94 at 2^{20} (Table 4). At 2^{20} rows the garbled verifier is smaller than the trace it verifies, 7.5 garbled bits per committed bit, or 38 kB per Keccak- f permutation. A verifier of fixed size, such as a Groth16 verifier, has η inversely proportional to the computation its proof covers as well; comparing η across proof systems therefore needs the same computation and cells of the same kind, bits here against field elements in a prime-field AIR.

Where the gates go. Table 5 breaks the 2^{18} circuit down by the verifier phase that emits each gate. The multi-STARK layers account for 70.8% and WHIR for 29.2%. Three of the largest items,

phase	non-free gates	share
ring switch closing	31,633,960	33.6%
WHIR Merkle paths	14,613,722	15.5%
column combination	11,582,352	12.3%
AIR constraints (1,650, Horner under α)	10,736,110	11.4%
STARK transcript	10,531,194	11.2%
WHIR leaf hashes	3,320,880	3.5%
WHIR closing query weights	3,069,960	3.3%
WHIR query folds	2,624,400	2.8%
WHIR transcript	1,603,940	1.7%
zerocheck sumcheck	1,227,336	1.3%
ring switch statement	835,688	0.9%
other (9 phases)	2,290,006	2.4%
total	94,069,548	100%

Table 5: A third of the 2^{18} -row verifier is the ring switch closing, and the five largest phases make up 84%. Non-free gates by the verifier phase that emits them.

the STARK transcript, the column combination and the AIR constraints, grow with the number of columns, so a narrower AIR shrinks them directly. The largest single item is the ring switch closing: 256 multiplications per packed variable, plus a fixed 11×512 for the column selector.

WHIR alone, and its parameters. Garbling only the WHIR opening of a 2^{18} -variable polynomial (rate 1/32, folding 4, $35 + 22 + 16$ queries) cost 24.3M non-free gates with single-root commitments, and 21.2M with a Merkle cap of 32 roots. That cap is Plonky3’s recommended height and was the optimum in our sweep of 1 to 256 roots. Three quarters of the WHIR circuit was hashing. Across rates 1/16–1/64 and folding factors 2–5, rate 1/32 with folding 4 gave the smallest circuit among the configurations we kept; rate 1/64 saved a further 10% of gates at the price of a prover domain twice as large.

Why a binary field. We also built KoalaBear ($p = 2^{31} - 2^{24} + 1$) and Poseidon2 on wires, to price the prime-field alternative. A KoalaBear multiplication is 3,451 AND gates, and a degree-4 extension multiplication 60,615, against 2,187 for all of $\mathbb{F}_{2^{128}}$. One Poseidon2 permutation of width 16 is 1,240,747 AND gates: 120 Blake3 compressions. The WHIR opening of a 2^{20} KoalaBear configuration (2,157 permutations, about 7,500 extension multiplications) would be about 3.1×10^9 gates with Poseidon2, more than the whole Boolean-garbled Groth16 verifier, and still $0.3\text{--}0.5 \times 10^9$ with Blake3, against our 21M for a 2^{18} -variable opening. The four-fold difference in size does not account for a gap of 15 to 150 $\times$: a garbled STARK verifier should use a binary field.

Ablation. Table 6 isolates two of our choices on the full verifier. Replacing our Blake3 adder with the original two-AND adder, with everything else unchanged, grew the circuit by 32% at 2^5 rows and 29% at 2^8 : the adder saves a quarter of the whole verifier, because Blake3 is in the transcript, the leaf hashes and every Merkle path. Building the same 2^5 -row verifier as a stored circuit produced exactly the same gates and wires, but took 12.4 GiB and 283s, against 0.69 GiB and 33s for the plan and the streamed garbling together, including the garbling itself.

On-chain cost. Table 7 prices a dispute at 2^{18} rows. The happy path, Kickoff and Take1, and the Challenge and Disprove transactions have constant size; BitVM3 reports its Disprove at 93 vB,

configuration	non-free gates	peak memory	time
2 ⁵ rows, this work (streamed)	44,573,121	0.69 GiB	33 s (plan + garble)
2 ⁵ rows, stored build	44,573,121	12.4 GiB	283 s (build only)
2 ⁵ rows, original Blake3 adder	58,997,590 (+32%)	–	–
2 ⁸ rows, this work	66,459,859	–	–
2 ⁸ rows, original Blake3 adder	85,768,603 (+29%)	–	–

Table 6: The one-AND adder removes about a quarter of the verifier’s gates, and streaming cuts memory 18×. Peak memory is the whole process’s, including proof generation, which is negligible at 2⁵ rows; the stored build’s time excludes garbling.

input scheme	vB/bit	measured, 1,041,024 bits			tuned	
		Assert	txs	fee	fee	assumption
Lamport (measured)	16.55	17.23 MvB	173	0.379 BTC (\$36,200)	\$29,400	hash (HASH160)
Antichain Winternitz (2, 16) [TF26]	11.31	11.77 MvB	118	0.259 BTC (\$24,700)	\$20,100	hash (HASH160)
Schnorr adaptor [WAA ⁺ 26, KTA ⁺ 26]	2.37	2.47 MvB	25	0.054 BTC (\$5,200)	\$4,200	discrete logarithm

references: BitVM2 unhappy path \$14,211 [GKS⁺26]; BitVM3 Assert of a 1,024-bit proof, 2.4 kvB [WAA⁺26]; BABE, all transactions, \$56.90 [GKS⁺26]

Table 7: Publishing the proof dominates a dispute: 2.5 MvB with adaptor signatures and 12 to 17 MvB with hash-based keys at 2¹⁸ rows. Assert size, the number of standard transactions (at most 100 kvB each) and the fee, at 2.2 sat/vB and \$95,500 per bitcoin as in BABE [GKS⁺26]. The Lamport rate is our measured script and witness with per-input overhead; the other rates are as their sources report them. The tuned column is the smallest input of Table 8.

and ours is the same hashlock spend [WAA⁺26]. Assert carries the proof and dominates: 2.47 MvB with adaptor signatures, and 11.8 to 17.2 MvB with hash-based keys. At the fee rate and price BABE uses [GKS⁺26], that is about \$5,200 for adaptor signatures, and \$24,700 to \$36,200 hash-based. BitVM2’s unhappy path cost \$14,211 in the same accounting [GKS⁺26]. A dispute over our verifier therefore costs about a third of BitVM2’s with adaptor signatures and up to 2.5× BitVM2’s without them, against tens of dollars for a Groth16 proof in BitVM3 or BABE [WAA⁺26, GKS⁺26]. The Assert transactions fill 2.5 to 17 blocks’ worth of space. Split into standard transactions of at most 100 kvB, they are 25 to 173 transactions, which must confirm within Δ_A ; BitVM3 uses a window of one week [WAA⁺26]. The challenge bond must cover this cost, which makes a challenge expensive for an honest challenger too.

How far the input shrinks. We counted the verifier’s input bits by proof component from the WHIR schedule alone. The count equals the measured input count of Table 4 at all five sizes. We then swept rates 1/8 to 1/256, folding factors 2 to 6 and grinding budgets up to 48 bits at 2¹⁸ rows, keeping 110 bits per WHIR term in the Johnson regime (Table 8). The smallest input, at rate 1/256, folding 4 and 48 bits of grinding, is 845,952 bits, 19% below the measured configuration, at 103.9 bits of composed soundness with every term assessed. Its WHIR part shrinks by 35%, but 480,256 bits do not depend on WHIR at all: the opened values are 416,000 bits, two rows of 1,625 columns at 128 bits each. The tuning also moves soundness from queries to grinding, which halves under quantum search (§5.1). Below this the input shrinks only with a narrower AIR, whose

component	measured	tuned
opened values (current and next row)	416,000	416,000
zerocheck and ring switch	64,256	64,256
WHIR Merkle paths	346,624	239,616
WHIR leaf rows	163,840	92,160
WHIR caps, OOD answers, sumchecks, final polynomial	50,304	33,920
total input bits	1,041,024	845,952
STIR queries per round	33 + 20 + 15 + 12	16 + 12 + 9 + 8
rate, folding, grinding budget	1/32, 4, 32	1/256, 4, 48
composed soundness (bits)	104.3	103.9

Table 8: Tuning WHIR removes a fifth of the input, and the AIR’s opened values set the floor. Input bits at 2^{18} rows by proof component, for the measured configuration and the smallest of the sweep; 128 bits per field element and 256 per digest, with full Merkle paths below each cap.

approach	statement	off-chain size	input bits	assumptions
Boolean garbled verifier [Bit25]	Groth16	42 GiB	1,019	pairings, trusted setup
native-group garbling [KFLT26]	Groth16	2.4 MiB	n/r	pairings, trusted setup
garbled DV-Pari (bitvm-gc, our measurement)	DV-SNARK	7.7 GB	n/r	elliptic curves, secret trapdoor
BABE [GKS ⁺ 26]	Groth16	22.2 MiB	508	pairings (GGM+RO), trusted setup
partial-binding WE [Dua26]	Groth16, late inputs	$(0.5\text{--}1.1) \times 10^9$ gates [†]	$508 + 254 D $	BABE + DLog
PIPE with AADP	pairing	≈ 338 TB (estimate)	one hash	heuristic
WE [ACG ⁺ 26, SRG ⁺ 26]	SNARK			
this work	STARK over $\mathbb{F}_{2^{128}}$	1.5 GB	1,041,024	hash functions[‡]

Table 9: Hash-only assumptions cost off-chain size and on-chain input. Off-chain artifact per instance, in bytes (garbled circuits at 16 bytes per non-free gate), and the number of input bits revealed on-chain at dispute time, as each source reports them; n/r: not reported. [†]Gate count as the note reports it, without a byte figure. [‡]The verifier’s assumption; the protocol’s pre-signed transactions, and adaptor-signature inputs if used, add discrete logarithms (§6). Our figures are for the 2^{18} -row Keccak- f AIR.

opened values scale with its width, or by recursion into a proof whose own WHIR opening is still some 0.37M bits at these parameters.

Comparison. Table 9 places the result among approaches to verifying proofs on Bitcoin (§8). For the designated-verifier pairing SNARK we garbled bitvm-gc’s circuit with our streaming garbler: 481.6M non-free gates and 1.92×10^9 wires, with at most 15.6M live, in 2.2 GB and under six minutes. The circuit accepted the valid proof and rejected it with one witness bit flipped; built as a stored circuit, the same verifier needed more than 64 GB. Our verifier has $5.1\times$ fewer non-free gates, and about $30\times$ fewer than the Boolean-garbled Groth16 verifier. Garbling Groth16 with native group operations [KFLT26] and witness encryption [GKS⁺26] produce far smaller artifacts, of megabytes, and their proofs need far fewer input bits; all of them rely on pairings and a trusted setup.

8 Related work

We group prior work by how it verifies a proof on Bitcoin: by garbling a verifier, by witness encryption, or through small pairing-based SNARKs that both approaches rely on. We close with the garbling techniques we use.

BitVM and garbled verifiers. BitVM2 [LAA⁺25] verified optimistically by bisecting the verifier’s trace on-chain. BitVM3 garbled the verifier instead [Lin25, Che25], and its formal treatment defines BitVM3-CORE as a sound and complete on-chain proof system with on-chain costs of about \$9 [WAA⁺26]. The Boolean-garbled Groth16 verifier [Bit25] has about 3×10^9 non-free gates, 42 GiB, with cut-and-choose setup of over three hours at $(N, M) = (181, 7)$ [GKS⁺26]. Garbling Groth16 with native elliptic-curve group operations reduced the garbled program to 2.4 MiB [KFLT26]. Glock [Eag25] formalized garbled locks and garbled a designated-verifier variant of Pari; BABE reports its garbled circuit at about 12 MB on a binary elliptic curve [GKS⁺26]. GOAT’s bitvm-gc [GOA26b] garbles Groth16 and a designated-verifier form of Pari [DMS24] over BN254 and over the binary curve sect233k1. We garble a full STARK verifier with the same garbling code, and we contribute our streaming garbler and Blake3 circuit back to it. BitVM3’s comparison of garbling schemes already lists a Yao-style garbled STARK verifier, at about 10 GB of off-chain storage and as an estimate [WAA⁺26, Table 9]; we build one and measure 1.5 GB per instance. TRAPGC-DV [FBF⁺26] verifies Groth16 by designated evaluation: the prover wraps its proof in an outer Groth16 proof under a per-session trapdoor SRS and garbles a small circuit that checks it in scalar form, with cut-and-choose and a universal-composability analysis. It too rests on pairings.

Garbling on Bitcoin: security and inputs. Futoransky, Larotonda and Barbara [FLB25] broke the authenticity of the RSA-based garbling proposed for BitVM3 [Lin25], of the label forward-propagation variant [FDZ25], and of linear adaptors: coprime public exponents let the evaluator forge labels by extended Euclid, and linear adaptors fall to the Franklin–Reiter related-message attack. Our hash-based garbling has no such label algebra. Mosaic [KTA⁺26] made cut-and-choose maliciously secure with an on-chain footprint independent of the number of garbled copies, and Antichain Winternitz [TF26] published input labels on-chain at up to 52.9% less cost than Lamport signatures. Both are independent of the garbled circuit; \$6 prices both for our million-bit input.

Verifying a STARK in Script. Our companion work, in the same repository as this paper’s code, verifies the same kind of proof directly in Bitcoin Script, with Poseidon2 over KoalaBear and WHIR and without OP_CAT [GOA26a]. It needs no garbling and no trusted party, but its 2^{20} example configuration is 1.24 GB of script, 309 blocks, and a standard transaction holds less than one Poseidon2 permutation, so it must be split into sub-query steps. Garbling moves that cost off-chain.

Witness encryption on Bitcoin. BABE [GKS⁺26] replaced the garbled Groth16 verifier with witness encryption for linear pairing relations [GKPW24]. It made the one quadratic pairing $e(\pi_1, \pi_2)$ linear by computing $r\pi_1$ in a small garbled circuit, built from decomposable randomized encodings on top of Argo MAC [EL26]. The per-instance artifact is 22 MiB, the on-chain cost is comparable to BitVM3, and cut-and-choose setup takes seconds. Partial-binding witness encryption [Dua26] extended BABE to public inputs known only at dispute time, with a dual-output

garbled circuit and Lamport-soldered inputs, relying on Groth16’s linear public-input aggregation.

Witness encryption for covenants. Bitcoin PIPEs v2 [ACG⁺26] witness-encrypt a Schnorr signing key under an NP statement to emulate covenants without a soft fork. They define witness signatures, and they map BitVM’s disprove condition to a relation $f(w) = 0 \wedge h = \text{Hash}(w)$, which requires the witness at setup. Their witness encryption comes from arithmetic affine determinant programs [SRG⁺26], an arithmetic variant of affine determinant programs [BIJ⁺20]. The ciphertext grows as $v(2g+1)^2$ field elements, cubic in the constraint count, which is about 338 TB for a purpose-built 14,371-constraint pairing-SNARK verifier. Security is heuristic: an earlier version fell to a rank-drop attack, and the linearization analysis is described as inconclusive. All witness-encryption designs above are smaller off-chain than a garbled verifier, but all require the statement to be a pairing SNARK, so a STARK would first have to be wrapped in one, and none is post-quantum.

Small SNARK verifiers. Pari [DMS24] has the smallest known pairing-based proofs, two $\mathbb{G}_1$ and two field elements, with a three-pairing verifier. Its equiefficient polynomial commitments also give Garuda, a prover with free linear gates. Pari underlies bitvm-gc’s designated-verifier SNARK, Glock, and the SNARK of the AADP construction; Glock and bitvm-gc’s `sect233k1` verifier use its designated-verifier form on a binary curve, which needs no pairing but still rests on discrete logarithms. Groth16 [Gro16] remains the deployed baseline.

Binary-field proof systems. Binius [DP25] builds succinct arguments over towers of binary fields, and ring switching [DP24] turns a claim about a polynomial’s bits into one about its packed extension-field form; Plonky3’s Boolean WHIR commitment, which we garble, uses both ideas with WHIR [ACFY25] as the proximity test. Its cheap additions are what make a garbled verifier small.

Garbling. We use privacy-free garbling [FNO15] with free-XOR [KS08], one ciphertext per AND gate [ZRE15], and the syntax of Bellare, Hoang and Rogaway [BHR12b]; free-XOR’s security rests on circular correlation robustness [CKKZ12]. Witness encryption in general goes back to Garg, Gentry, Sahai and Waters [GGSW13].

9 Discussion

Why the operator garbles. The roles could be swapped, with a challenger garbling and the operator evaluating its own proof to obtain an accept label that unlocks its payment. The operator then needs the input labels of its proof from the garbler. Publishing both labels of every input wire would disclose Δ , their XOR, and with it every label; handing them over on request lets the garbler withhold them, so that the operator’s payment depends on the challenger’s cooperation. With the operator as garbler, a withheld label only hurts the party that withholds it.

The input is the cost. The protocol is enforceable on today’s Bitcoin, and its happy path costs as little as any BitVM-style protocol. A dispute, however, costs what the proof’s size dictates, which is a thousand times the input of a Groth16 proof (Table 7). A dispute occurs only when the operator is challenged, and the bond puts its cost on the challenger. Tuning the proof removes a fifth of the input (Table 8); the rest needs a narrower AIR for the statement actually verified, or recursion into a proof of a small verifier AIR whose opened values are few. Hash-based keys cost 5 to $7\times$ adaptor signatures on-chain, the price of a protocol whose input layer a quantum adversary does not break.

Other AIRs. Our sizes are for the 1,625-column Keccak AIR. Plonky3 also has binary AIRs for SHA-256 and Blake3, but they read no next-row columns, which our verifier currently assumes; supporting them changes the opening layout and is left to future work. The column-dependent terms, about a third of the circuit, scale with the AIR’s width.

Verifying the garbling. A garbler must be kept honest, and in current deployments that is cut-and-choose over many instances. Its cost is linear in the garbled circuit. At $(N, M) = (181, 7)$ and 2^{18} rows, the operator garbles 181 instances at about 57s each, 2.9 core-hours or about 11 minutes on 16 cores, and a challenger stores the 7 kept instances, 10.5 GB. BitVM3 reports 3 hours 24 minutes on 16 cores and 288.4 GB for the Boolean-garbled Groth16 verifier at the same parameters [WAA⁺26]. Against native-group garbling of Groth16 [KFLT26] or BABE, whose artifacts are megabytes, our setup is larger. The streaming garbler re-garbles an opened instance from its seed in one garbling pass, reusing the plan that all instances share. These setup figures are derived from the measured garbling time, not measured end to end.

Further reductions. The ring switch closing is a third of the circuit and is the first target. An $\mathbb{F}_{2^{128}}$ multiplier below Karatsuba’s 3^7 is a second: bitvm-gc’s $\mathbb{F}_{2^{233}}$ multiplier, which evaluates, multiplies and interpolates over $\mathbb{F}_{2^9}$, costs 2,097 AND gates for a field almost twice as wide, at the price of 1.5M XORs per product. Per-tree Merkle cap heights and a narrower AIR for the statement actually verified reduce the hashing and column terms.

10 Conclusion

We have shown that a STARK verifier over a binary tower field fits a garbled circuit well: additions are free, multiplication is cheap, and the remaining cost is hashing, which we have made twice as cheap. We have garbled the complete Plonky3 multi-STARK verifier for a 2^{18} -row Keccak trace in 94.1M non-free gates and 1.5 GB, about $30\times$ fewer non-free gates than the Boolean-garbled Groth16 verifier, with no pairing, no trusted setup and no number-theoretic assumption. We have also shown what post-quantum security costs: the parameters measured offer about 52 bits against a quantum adversary, and 100 bits need 256-bit labels and a 256-bit field for an estimated $3\times$ the gates. On today’s Bitcoin the verifier fits a BitVM3-style protocol whose happy path is cheap, but whose dispute publishes a million-bit proof, 2.5 to 17 MvB depending on the input keys; tuning the proof removes a fifth of it. Two steps remain before deployment: measuring the post-quantum parameters, and a proof whose opened values, and hence its on-chain input, are small.

References

- [ACFY24] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. STIR: Reed–solomon proximity testing with fewer queries. In *Advances in Cryptology – CRYPTO 2024*, 2024. ePrint 2024/390.
- [ACFY25] Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, and Eylon Yogev. WHIR: Reed–Solomon proximity testing with super-fast verification. In *EUROCRYPT 2025*, 2025. Cryptology ePrint Archive, Paper 2024/1586.
- [ACG⁺26] Michel Abdalla, Brent Carmer, Muhammed El Gebali, Handan Kilinc-Alper, Mikhail Komarov, Yaroslav Rebenko, Lev Soukhanov, Erkan Tairi, Elena Tatumova,

and Patrick Towa. Bitcoin PIPEs v2: Covenants and ZKPs on Bitcoin via witness encryption. Cryptology ePrint Archive, Paper 2026/186, 2026.

- [BHR12a] Mihir Bellare, Viet Tung Hoang, and Phillip Rogaway. Adaptively secure garbling with applications to one-time programs and secure outsourcing. In *ASIACRYPT 2012*, volume 7658 of *LNCS*, pages 134–153, 2012.
- [BHR12b] Mihir Bellare, Viet Tung Hoang, and Phillip Rogaway. Foundations of garbled circuits. In *ACM CCS 2012*, pages 784–796, 2012.
- [BHT98] Gilles Brassard, Peter Høyer, and Alain Tapp. Quantum cryptanalysis of hash and claw-free functions. In *LATIN 1998*, volume 1380 of *LNCS*, pages 163–169, 1998.
- [BIJ⁺20] James Bartusek, Yuval Ishai, Aayush Jain, Fermi Ma, Amit Sahai, and Mark Zhandry. Affine determinant programs: A framework for obfuscation and witness encryption. In *ITCS 2020*, volume 151 of *LIPIcs*, pages 82:1–82:39, 2020.
- [Bit25] BitVM. Garbled SNARK verifier.
<https://github.com/BitVM/garbled-snark-verifier>, 2025.
- [Can89] David G. Cantor. On arithmetical algorithms over finite fields. *Journal of Combinatorial Theory, Series A*, 50(2):285–300, 1989.
- [CBBZ23] Binyi Chen, Benedikt Bünz, Dan Boneh, and Zhenfei Zhang. HyperPlonk: Plonk with linear-time prover and high-degree custom gates. In *Advances in Cryptology – EUROCRYPT 2023*, 2023. ePrint 2022/1355.
- [Che25] Weikeng Chen. SoK: BitVM with succinct on-chain cost. Cryptology ePrint Archive, Paper 2025/1253, 2025.
- [CKKZ12] Seung Geol Choi, Jonathan Katz, Ranjit Kumaresan, and Hong-Sheng Zhou. On the security of the “free-XOR” technique. In *TCC 2012*, volume 7194 of *LNCS*, pages 39–53, 2012.
- [CMS19] Alessandro Chiesa, Peter Manohar, and Nicholas Spooner. Succinct arguments in the quantum random oracle model. In *TCC 2019*, 2019. Cryptology ePrint Archive, Paper 2019/834.
- [DMS24] Michel Dellepère, Pratyush Mishra, and Alireza Shirzad. Garuda and pari: Faster and smaller SNARKs via efficient polynomial commitments. Cryptology ePrint Archive, Paper 2024/1245, 2024.
- [DP24] Benjamin E. Diamond and Jim Posen. Polylogarithmic proofs for multilinear over binary towers. Cryptology ePrint Archive, Paper 2024/504, 2024.
- [DP25] Benjamin E. Diamond and Jim Posen. Succinct arguments over towers of binary fields. In *Advances in Cryptology – EUROCRYPT 2025*, 2025. ePrint 2023/1784.
- [Dua26] Stephen Duan. Partial-binding witness encryption over Groth16.
GOATNetwork/bitvm-gc, docs/partial_binding_we.tex, 2026.
- [Eag25] Liam Eagen. Glock: Garbled locks for Bitcoin. Cryptology ePrint Archive, Paper 2025/1485, 2025.

- [EL26] Liam Eagen and Ying Tong Lai. Argo MAC: Garbling with elliptic curve MACs. Cryptology ePrint Archive, Paper 2026/049, 2026.
- [FBF⁺26] Ariel Futoransky, Fadi Barbàra, Ramses Fernandez, Gabriel Larotonda, and Sergio Demian Lerner. TRAPGC-DV: Trapdoor garbled circuit designated verification of Groth16 on Bitcoin. Cryptology ePrint Archive, Paper 2026/1430, 2026.
- [FDZ25] Alva Fu, Stephen Duan, and Ethan Zhu. Instantiating BitVM3 from label forward propagation. <https://www.goat.network/bitvm3-label-forward-propagation>, 2025.
- [FLB25] Ariel Futoransky, Gabriel Larotonda, and Fadi Barbara. A note on the security of the BitVM3 garbling scheme. Cryptology ePrint Archive, Paper 2025/1291, 2025.
- [FNO15] Tore Kasper Frederiksen, Jesper Buus Nielsen, and Claudio Orlandi. Privacy-free garbled circuits with applications to efficient zero-knowledge. In *EUROCRYPT 2015, Part II*, volume 9057 of *LNCS*, pages 191–219, 2015.
- [GGSW13] Sanjam Garg, Craig Gentry, Amit Sahai, and Brent Waters. Witness encryption and its applications. In *45th ACM STOC*, pages 467–476, 2013.
- [GKPW24] Sanjam Garg, Dimitris Kolonelos, Guru-Vamsi Policharla, and Mingyuan Wang. Threshold encryption with silent setup. In *CRYPTO 2024, Part VII*, volume 14926 of *LNCS*, pages 352–386, 2024.
- [GKS⁺26] Sanjam Garg, Dimitris Kolonelos, Mikhail Sergeevitch, Srivatsan Sridhar, and David Tse. BABE: Verifying proofs on Bitcoin made 1000x cheaper. Cryptology ePrint Archive, Paper 2026/065, 2026.
- [GOA26a] GOAT Network. bitcoin-stark-verifier: a STARK verifier in Bitcoin Script without OP_CAT. <https://github.com/GOATNetwork/bitcoin-stark-verifier>, crates poseidon2 and whirl, 2026.
- [GOA26b] GOAT Network. bitvm-gc: garbled SNARK verifier and BABE for BitVM. <https://github.com/GOATNetwork/bitvm-gc>, 2026.
- [Gro96] Lov K. Grover. A fast quantum mechanical algorithm for database search. In *28th ACM STOC*, pages 212–219, 1996.
- [Gro16] Jens Groth. On the size of pairing-based non-interactive arguments. In *EUROCRYPT 2016, Part II*, volume 9666 of *LNCS*, pages 305–326, 2016.
- [KFLT26] Nakul Khambhati, Aaron Feickert, Christian Lewe, and Mukesh Tiwari. Garbling Groth16 with native group operations. Cryptology ePrint Archive, Paper 2026/2100, 2026.
- [KS08] Vladimir Kolesnikov and Thomas Schneider. Improved garbled circuit: Free XOR gates and applications. In *ICALP 2008, Part II*, volume 5126 of *LNCS*, pages 486–498, 2008.
- [KTA⁺26] Nakul Khambhati, Mukesh Tiwari, Azz, Sapin Bajracharya, Manish Bista, Liam Eagen, Christian Lewe, and Aaron Feickert. Mosaic: Practical malicious security for garbled circuits on Bitcoin. Cryptology ePrint Archive, Paper 2026/812, 2026.

- [LAA⁺25] Robin Linus, Lukas Aumayr, Zeta Avarikioti, Matteo Maffei, Andrea Pelosi, Orfeas Stefanos Thyfronitis Litos, Christos Stefo, David Tse, and Alexei Zamyatin. Bridging Bitcoin to second layers via BitVM2. Cryptology ePrint Archive, Paper 2025/1158, 2025.
- [LCH14] Sian-Jheng Lin, Wei-Ho Chung, and Yung-Hsiang S. Han. Novel polynomial basis and its application to Reed–Solomon erasure codes. In *IEEE 55th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 316–325, 2014.
- [LFKN92] Carsten Lund, Lance Fortnow, Howard Karloff, and Noam Nisan. Algebraic methods for interactive proof systems. *Journal of the ACM*, 39(4):859–868, 1992.
- [Lin25] Robin Linus. BitVM3: Efficient computation on Bitcoin. <https://bitvm.org/bitvm3-rsa.pdf>, 2025.
- [Nat15] National Institute of Standards and Technology. SHA-3 standard: Permutation-based hash and extendable-output functions. FIPS PUB 202, NIST, 2015.
- [OANWO20] Jack O’Connor, Jean-Philippe Aumasson, Samuel Neves, and Zooko Wilcox-O’Hearn. BLAKE3: One function, fast everywhere. <https://github.com/BLAKE3-team/BLAKE3-specs>, 2020.
- [Plo26] Plonky3 contributors. Plonky3: binary tower fields, WHIR and multi-STARK. <https://github.com/Plonky3/Plonky3>, 2026.
- [Sho94] Peter W. Shor. Algorithms for quantum computation: Discrete logarithms and factoring. In *35th FOCS*, pages 124–134, 1994.
- [SRG⁺26] Lev Soukhanov, Yaroslav Rebenko, Muhammad El Gebali, Mikhail Komarov, Handan Kilinc-Alper, Michel Abdalla, and Patrick Towa. Implementable witness encryption from arithmetic affine determinant programs. Cryptology ePrint Archive, Paper 2026/175, 2026.
- [TF26] Mukesh Tiwari and Aaron Feickert. Antichain Winternitz: Guaranteed garbled-circuit label revelation on Bitcoin with permissionless recovery. Cryptology ePrint Archive, Paper 2026/1568, 2026.
- [WAA⁺26] Robin Linus Woll, Ioannis Alexopoulos, Lukas Aumayr, Zeta Avarikioti, Matteo Maffei, and David Tse. BitVM3: Efficient Bitcoin bridges via garbled circuits. Cryptology ePrint Archive, Paper 2026/933, 2026.
- [Wie88] Doug Wiedemann. An iterated quadratic extension of GF(2). *The Fibonacci Quarterly*, 26(4):290–295, 1988.
- [ZKM26] ZKM. Hash-based is not the same as post-quantum. <https://www.zkm.io/blog/hash-based-is-not-the-same-as-post-quantum>, 2026.
- [ZRE15] Samee Zahur, Mike Rosulek, and David Evans. Two halves make a whole: Reducing data transfer in garbled circuits using half gates. In *EUROCRYPT 2015, Part II*, volume 9057 of *LNCS*, pages 220–250, 2015.